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# Rule-Based System
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if facts:
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```
    apply_rules()
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```
    infer_new_facts()
```

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else:
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```
    check_goal()
```

RULE-BASED SYSTEM

FORWARD CHAINING & BACKWARD CHAINING

A PROJECT ON KNOWLEDGE REPRESENTATION

FORWARD CHAINING (DATA-DRIVEN)



BACKWARD CHAINING (GOAL-DRIVEN)



LOGIC • RULES • REASONING • INFERENCE



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COURSE

DSAI 104 Knowledge Representation



1) Introduction:

- *Rule-based systems are AI systems that use a set of facts and logical rules to infer new information.
This project implements two inference techniques:
Forward Chaining and Backward Chaining.*
- *Forward Chaining starts from known facts and applies rules to infer new facts.
Backward Chaining starts from a goal and works backward to determine whether the goal can be achieved.*

2) Project Objective:

- *The objective of this project is to design and implement a rule-based system using Python.*
- *The system applies logical IF-THEN rules to a knowledge base in order to infer conclusions using Forward Chaining and Backward Chaining algorithms.*

3) Knowledge Base:

Initial Facts:

- *fever*
 - *cough*
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4) Rules:

Rule Number	Rule
Rule 1	IF fever AND cough THEN flu
Rule 2	IF flu THEN rest needed
Rule 3	IF rest needed THEN stay home

5) Forward Chaining:

- Forward Chaining is a data-driven inference technique.
 - The system starts with known facts and repeatedly applies rules whose conditions are satisfied.
 - New facts are added until no more rules can be applied.
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6) Execution Trace

Step	Rule Checked	Condition Status	Action
1	IF cough AND fever THEN flu	Satisfied	flu inferred
2	IF flu THEN rest_needed	Satisfied	rest_needed inferred
3	IF rest_needed THEN stay_home	Satisfied	stay_home inferred
4	IF cough AND fever THEN flu	Already exists	No new fact
5	IF flu THEN rest_needed	Already exists	No new fact
6	IF rest_needed THEN stay_home	Already exists	No new fact

The system successfully inferred all possible facts from the initial knowledge base.

7) Final facts:

Final Facts
fever
cough
flu
rest_needed
stay_home

8) Backward Chaining:

- *Backward Chaining is a goal-driven inference technique.*
- *The system starts with a goal and searches for rules that can satisfy this goal.*
- *Sub-goals are generated recursively until reaching known facts.*

9) Execution Trace

Step	Current Goal	Rule Used	Result
1	stay_home	rest_needed $\rightarrow$ stay_home	Sub-goal generated
2	rest_needed	flu $\rightarrow$ rest_needed	Sub-goal generated
3	flu	cough + fever $\rightarrow$ flu	Sub-goals checked
4	cough	Found in facts	Success
5	fever	Found in facts	Success
6	flu	Goal achieved	Success
7	rest_needed	Goal achieved	Success
8	stay_home	Goal achieved	Success

The backward chaining process started with the goal "stay_home" and recursively checked all required sub-goals until reaching the known facts.

10) Final Goal Status:

Goal	Status
cough	Achieved
fever	Achieved
flu	Achieved
rest_needed	Achieved
stay_home	Achieved

11) Comparison Between Forward and Backward Chaining:

Forward Chaining	Backward Chaining
Starts with facts	Starts with goal
Data-driven	Goal-driven
Infers all possible facts	Checks only required goals
Moves from facts to conclusion	Moves from goal to facts
Uses iterative process	Uses recursive process

Forward chaining begins with known facts and generates all possible conclusions, while backward chaining starts with a specific goal and works backward to verify whether the goal can be achieved.

12) Advantages of Rule-Based Systems:

- Easy to understand and implement.
- Provides clear logical reasoning.
- Useful in expert systems and decision-making.
- Easy to debug and trace.
- Supports knowledge representation.

13) Limitations:

- *Performance decreases with very large rule sets.*
 - *Rules must be manually defined.*
 - *Difficult to handle uncertainty.*
 - *Complex systems may require many rules.*
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14) Conclusion:

- *This project demonstrated the implementation of a rule-based system using Forward Chaining and Backward Chaining techniques.*
 - *The system successfully applied logical IF-THEN rules to infer conclusions from the knowledge base.*
 - *Forward chaining was used to infer new facts from initial facts, while backward chaining was used to verify goals through recursive reasoning.*
 - *The project also demonstrated the reasoning process step-by-step using execution traces.*
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